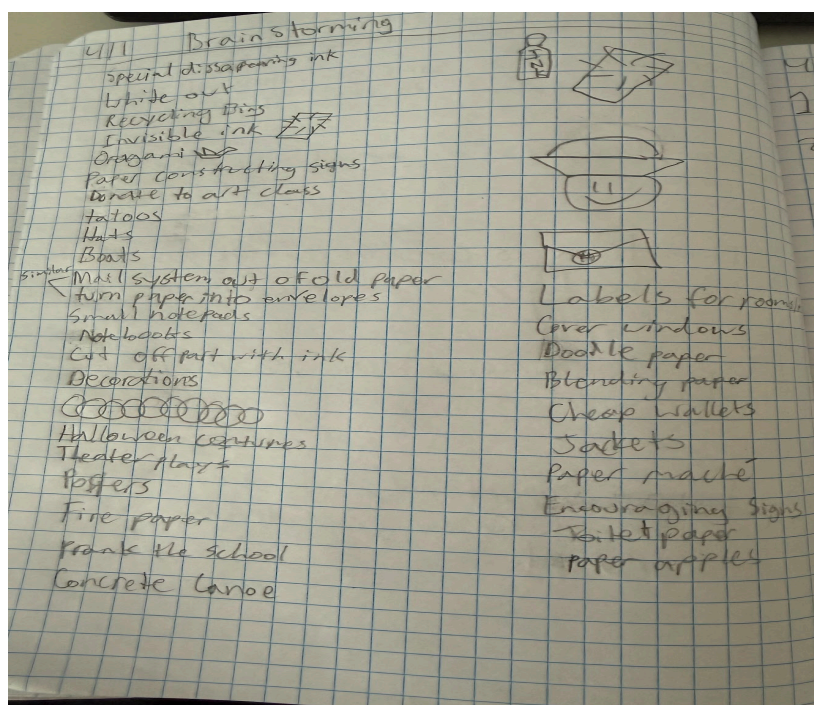
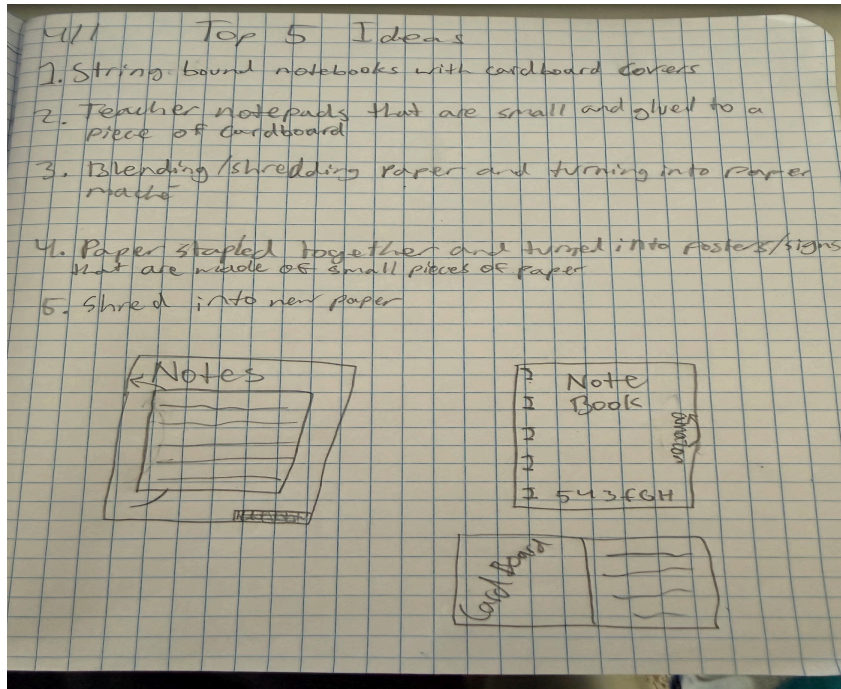


Element D: Design Concept Generation, Analysis, and Selection

We are making a solution to reduce the paper waste. This is my team and I's process for finding our solution. First, my teammates and I came up with as many ideas as we could in ten minutes individually. We then met up as a team and wrote down everybody's ideas. After hearing everybody's ideas we narrowed them down to the five best ideas. Once we had all five ideas, we typed them and our design requirements on a decision matrix. We weighed our safety requirements more because it was the highest priority. We used the matrix to figure out which one was the best. We landed on our final solution and my team and I were happy with it. Our final solution is a notepad made out of misprinted paper and it rests on a chipboard base. We chose it because it was the highest on the decision matrix, and we thought it would be the most realistic to make. It seems like it will meet all the design requirements other than possibly price. We weighed safety because we agreed it was the highest priority, and there seems to be no safety hazards in any process with the notepads.



Brainstorming process



Top ideas and sketches

Team: The Big Apples

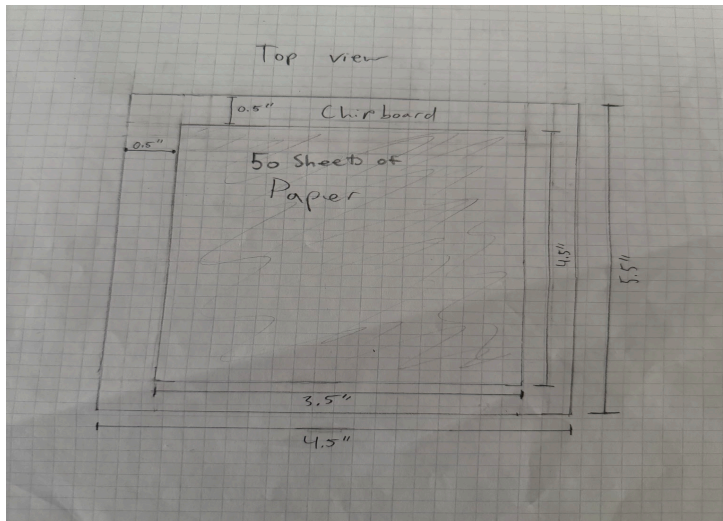
Design Requirements	Proposed Solutions					
	Current Solution: Notes on misprinted paper	Idea #1: String bound notebooks, reuse cardboard covers	Idea #2: Teacher notepads, glued to cardboard, small	Idea #3: Shredding paper & turning it into paper mache	Idea #4: Paper stapled together into posters	Idea #5: Shredding paper & turning it into new paper
Safe process, no sharp objects/edges, final product is safe to touch - (High) X2	0	0	0	-2	-2	-2
Reuse 90% of scrap paper from misprints and notebooks - (High)	0	+1	+1	+1	+1	+1
One side of each paper must be blank - (Med)	0	0	0	+1	0	+1
8.5 x 11 in. or smaller - (Med)	0	0	+1	0	-1	0
Less than \$2.00 per solution - (Low)	0	-1	-1	-1	-1	-1
at least 50 sheets of paper - (Low)	0	+1	+1	+1	+1	+1
	0	0	0	0	0	0
	0	0	0	0	0	0
	0	0	0	0	0	0

Total Score	0	+1	+2	0	-2	0
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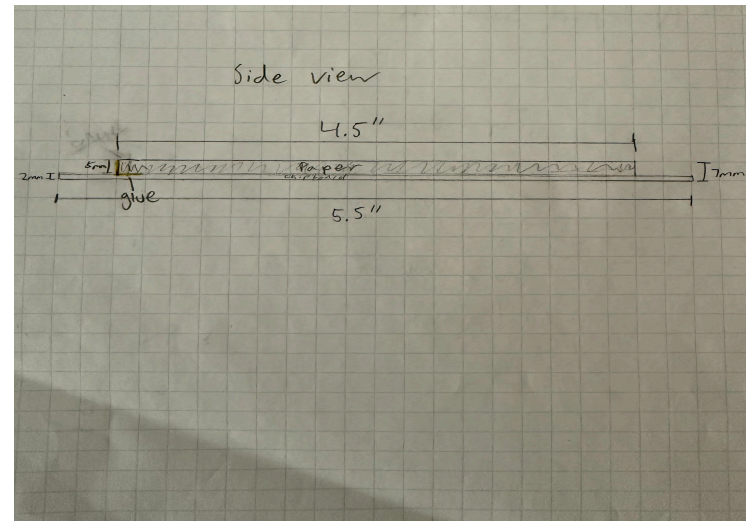
Scoring
 0 = same as current solution
 +1 = better than current solution
 -1 = worse than current solution

Decision Matrix

We would take our chip board and cut it to the correct dimensions. We would take a normal size of misprinted paper and cut it twice. We would take 50 pieces of paper and use a PVA (polyvinyl acetate) glue to hold the paper together. We would use the same glue to connect the notepad and chipboard together.



Top View



Side View

The dimensions are 4.5 inches by 5.5 inches for the chipboard. We will take the paper and position it 0.5 inches away from all edges. The paper should be 3.5 inches by 4.5 inches. The chipboard is 2mm thick, and the 50 pieces of paper are 5mm. The total thickness of the product is 7mm. The materials and tools we are using are PVA glue, chipboard, paper, a paper guillotine, scissors, a ruler, a pencil, clips, cardboard, and a paint brush. The PVA glue will hold the paper together and allow it to stick to the chipboard. The glue costs \$10, and the chipboard costs \$18. The equivalent cost of materials is \$18, but we hope that each notepad costs less than two dollars to build. We can take the averages of how much paper we use to cover the price, and we can count how many pieces of paper we use per solution to figure out how many sheets of paper are being reused.